

Project Overview

This project is a web-based Rental Management System that allows users to manage rental properties, tenants, and rental payments. The system is built using Flask for backend processing, SQLite for database management, and HTML for the front-end interface.

In this version (Assignment 3), we have implemented the following key features:

- Database Setup for Users, Properties, and Messages: Established the databases for managing user accounts, property listings, and communication messages.
- Enhanced Functionality for User Registration, Login, and Logout: Improved the overall user experience during registration, login, and logout processes.
- Property Listings and Tenant Messaging: Tenants can leave messages for landlords on publicly available properties, making it easier for tenants to inquire about listings.
- Property Creation and Availability Management: Landlords can create new property listings and specify available rental periods.
- Direct Chat Feature: Implemented a chat feature that allows landlords and tenants to communicate directly, enhancing interaction efficiency.

Project Structure

```
/Assignment-3
  /HouseRentSystem
    /templates          # HTML
    app.py              # Entrance
    rental_management.db # Database
    tests               # Pytest
    requirements
    readme-first.pdf
    screenshots.jpg
    task-distribution.pdf
```

How-To-Run

1. Make sure you are in the root directory of the project (where `requirements.txt` and `app.py` are located) before running the following commands.
2. In the root directory, create a virtual environment by running the following command:

```
python3 -m venv venv
```

Here the second `venv` is the name of the virtual environment. You can use any name you want.

3. Activate the virtual environment by running the following command:

```
source venv/bin/activate
```

If you are using Windows, you can activate the virtual environment by running the following command:

venv\Scripts\activate

4. Install the required packages by running the following command:

pip install -r requirements.txt

5. Run the following command to start the server:

python app.py

How-To-Run-Test-Scripts (pytest)

1. Make sure the virtual environment is activated:

On macOS/Linux:

Source venv/bin/activate

On Windows:

venv\Scripts\activate

2. Navigate to the root directory of the project (if not already there).
3. Run the following command to execute the test scripts:
pytest
4. The pytest command will automatically find and run all the test scripts located in the tests folder.
5. Once the tests have been run, the terminal will display the results, showing which tests have passed or failed.